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RE: Staff AI Software Engineer, Qualcomm Canada (Job ID 3085356)

Dear Hiring Team,

I am writing to apply for the Staff AI Software Engineer position with Qualcomm Canada in Markham. I am drawn to the opportunity to build the high-performance software, profiling tools, and framework enhancements behind on-device neural-network execution. My background combines staff-level software ownership, Python-based AI systems, compiler/runtime research, large-scale performance telemetry, and cross-functional delivery in highly regulated production environments.

At RBC, I architected an OpenTelemetry platform using Vector, Fluent Bit, Kafka, and Logstash that ingests 150 TB of telemetry per day across more than 10,000 systems. I standardized diagnostic schemas and pipelines adopted by 20+ engineering teams, giving developers consistent tools for profiling, distributed tracing, latency SLOs, anomaly detection, and rapid root-cause analysis. Designing for throughput, bounded resource usage, reliability, and vendor-neutral execution paths has made performance measurement and debugging central to how I build software.

I am also building ASB Assist, an LLM + RAG capability with retrieval and inference pipelines, offline evaluation, regression testing, grounding controls, and audit trails before production rollout. Earlier, as HealthCard's first engineer, I integrated transformer-based document classification into medical and insurance workflows and deployed monitored inference APIs on AWS EKS, reducing manual review by 60%. These projects have required careful evaluation of runtime behavior, accuracy, latency, operational signals, and release quality around neural-network workloads.

My lower-level systems foundation includes research at Dalhousie University on call-stack decision algorithms and Python tracing tools for compiler/runtime analysis. I have continued developing that performance-oriented perspective through a personal edge-inference benchmarking prototype that compares PyTorch and TensorFlow model latency, throughput, memory usage, numerical accuracy, and quantization tradeoffs across constrained configurations. While my recent production work has centered on Python, Go, and TypeScript rather than commercial C++ development, I bring strong algorithms, debugging, Linux, distributed-systems, and runtime-analysis fundamentals, together with the curiosity to deepen the hardware-specific C++ and accelerator skills required by Qualcomm's AI software stack.

As a hands-on staff engineer, I lead five engineers through design, implementation, code reviews, testing, release readiness, and operational handoff while aligning product, security, risk, platform, and architecture teams. I would welcome the opportunity to bring that ownership to Qualcomm: optimizing neural-network execution, strengthening profiling and debugging workflows, partnering with hardware and customer teams, and shipping cohesive, commercial-quality AI software. Thank you for your consideration.

Best regards,
Dhruv Doshi